

THE $1 \leq p \leq 2$ TRANSLATE-FRAME GAP AFTER FREEMAN–ODELL–SCHLUMPRECHT–ZSAK

FA.BANACH.001 LITERATURE-STATUS PACKET

STATUS

This is a literature-already-answered packet. The source paper is Freeman–Odell–Schlumprecht–Zsak, *Unconditional structures of translates for $L_p(\mathbb{R}^d)$* , arXiv:1209.4619. In Section 3, Theorem 3.2, the authors prove that if $2 < p < \infty$, $d \in \mathbb{N}$, and (λ_n) is any unbounded sequence in $\mathbb{R}^d$, then there is an unconditional Schauder frame

$$(T_{\lambda_n} f, g_n^*)_{n \in \mathbb{N}}$$

for $L_p(\mathbb{R}^d)$. This leaves the complementary arbitrary-translate frame case $1 < p \leq 2$ as the missing half of the dichotomy.

ANSWER SOURCE

Lev–Tselishchev, *There are no unconditional Schauder frames of translates in $L^p(\mathbb{R})$* , $1 \leq p \leq 2$, arXiv:2312.01757, explicitly identify this gap. Their introduction states that the existence of an unconditional Schauder frame of translates in $L^p(\mathbb{R})$, $1 < p \leq 2$, remained open after the Freeman–Odell–Schlumprecht–Zsak positive result for $p > 2$. Their Theorem 1.1 proves that, for $1 \leq p \leq 2$, there is no unconditional Schauder frame of the form

$$(T_{\lambda_n} g, g_n^*)_{n \in \mathbb{N}}$$

in $L^p(\mathbb{R})$.

IDENTIFICATION

Combining the two papers gives the sharp line-case dichotomy for unconditional Schauder frames whose vector side consists of translates of a single function:

existence exactly for $2 < p < \infty$.

The supporting authors explicitly know they are addressing the remaining $1 < p \leq 2$ case after arXiv:1209.4619, so this is an already-known literature answer, not a new proof from this run.

SCOPE

This does not settle the long-standing Schauder-basis problem for translates in $L_p(\mathbb{R})$. It also does not settle the final Section 6 questions in arXiv:1209.4619 about conditional bases/basic sequences or blocking into unconditional FDDs. The resolved item is the unconditional Schauder translate-frame gap for $1 \leq p \leq 2$.

REFERENCES

REFERENCES

- [1] D. Freeman, E. Odell, Th. Schlumprecht, and A. Zsák, *Unconditional structures of translates for $L_p(\mathbb{R}^d)$* , arXiv:1209.4619.
- [2] N. Lev and A. Tselishchev, *There are no unconditional Schauder frames of translates in $L^p(\mathbb{R})$, $1 \leq p \leq 2$* , arXiv:2312.01757.